

torch.repeat_interleave#

https://pytorch.org/docs/stable/generated/torch.repeat_interleave.html#torch

Description:

Repeat elements of a tensor. This is different from `torch.Tensor.repeat()` but similar to `numpy.repeat`. `input` (Tensor) – the input tensor. `repeats` (Tensor or int) – The number of repetitions for each element. `repeats` is broadcasted to fit the shape of the given axis.

Function Signature

```
torch.repeat_interleave(input, repeats, dim=None, *,  
output_size=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Return type

Parameters

Keyword

`output_size` (int, optional) – Total output size for the given axis (e.g. sum of repeats). If given, it will avoid stream synchronization needed to calculate output shape of the tensor.

Returns:

Tensor

Code Examples

```
>>> x = torch.tensor([1, 2, 3])
>>> x.repeat_interleave(2)
tensor([1, 1, 2, 2, 3, 3])
>>> y = torch.tensor([[1, 2], [3, 4]])
>>> torch.repeat_interleave(y, 2)
tensor([1, 1, 2, 2, 3, 3, 4, 4])
>>> torch.repeat_interleave(y, 3, dim=1)
tensor([[1, 1, 1, 2, 2, 2],
        [3, 3, 3, 4, 4, 4]])
>>> torch.repeat_interleave(y, torch.tensor([1, 2]), dim=0)
tensor([[1, 2],
        [3, 4],
        [3, 4]])
>>> torch.repeat_interleave(y, torch.tensor([1, 2]), dim=0,
output_size=3)
tensor([[1, 2],
        [3, 4],
        [3, 4]])
```

```
>>> torch.repeat_interleave(torch.tensor([1, 2, 3]))
tensor([0, 1, 1, 2, 2, 2])
```

Notes & Warnings

WARNING

Warning This is different from `torch.Tensor.repeat()` but similar to `numpy.repeat`.

Detailed Documentation

Documentation

Repeat elements of a tensor.

This is different from `torch.Tensor.repeat()` but similar to `numpy.repeat`.

`input (Tensor)` – the input tensor.

`repeats (Tensor or int)` – The number of repetitions for each element. `repeats` is broadcasted to fit the shape of the given axis.

`dim (int, optional)` – The dimension along which to repeat values. By default, use the flattened input array, and return a flat output array.

`output_size (int, optional)` – Total output size for the given axis (e.g. sum of repeats). If given, it will avoid stream synchronization needed to calculate output shape of the tensor.

Repeated tensor which has the same shape as input, except along the given axis.

If the `repeats` is `tensor([n1, n2, n3, ...])`, then the output will be `tensor([0, 0, ..., 1, 1, ..., 2, 2, ..., ...])` where 0 appears `n1` times, 1 appears `n2` times, 2 appears `n3` times, etc.

Repeats 0 `repeats[0]` times, 1 `repeats[1]` times, 2 `repeats[2]` times, etc.

`repeats (Tensor)` – The number of repetitions for each element.

Repeated tensor of size `sum(repeats)`.

